DISCRETE MATHEMATICS (BCSC 1010)

Assignment-1

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Set, Relation & Functions

Ques.1 Social Media Connections

Let U be the set of users. The relation R is defined as. $R = \{(a,b) \in U \times U \mid a \text{ is friend with } b\}$.

Reflexive: No, Usually, a user is not considered their own "friend" in social graph $(a,a) \notin R$.

Symmetric - Yes, if user a is friends with b, then b is friends with a, Therefore $(a,b) \in R \Rightarrow (b,a) \in R$.

Transitive - No, if a is friend with b and b friend with c, a is not necessarily friend with c. They are "friends of friends".

$$(a,b) \in R \;\&\; (b,c) \in R \not\Rightarrow (a,c) \in R.$$

Ques.2. Course Enrollment

$$f: S \rightarrow P(C)$$

where S is the set of students and P(C) is the power set of all courses.

Injective - (One to One): No. Multiple students can enroll in the exact same set of courses (e.g. two classmates in the same branch).

Surjective - (Onto): No. There are likely many combinations of courses in the power set that no student has actually selected.

Ques.3. Transportation Routes.

Modelling: This is a Directed Relation. if there is a bus from stop A to stop B, it is represented as an ordered pair (A, B).

Properties:

⇒ It is a Relation, not a function, because one bus stop (domain) can map to multiple destination stop (codomain).

⇒ It is Non-Symmetric if the bus route is one-way (e.g loop)

Ques.4. Equivalence Relation in Programming.

Definition: A relation that is Reflexive, Symmetric and Transitive.

Example: In a Java compiler, "Variable Data Type" is an equivalence Relation.

$Var_A \sim Var_B$ if both are of type Int

Optimization:

The system can partition data into Equivalence Classes. This allows the program to allocate a specific memory block for all int variables together, rather than treating each individually, which speeds up processing.

Q.5. Database Relations

Concept: A relationship between two tables (e.g. Students and Hostels) is a subset of the cartesian product of the two tables.

Functionality: if every student lives in exactly one hostel, the relationship is a function.

Efficiency: The mathematical mapping allows for indexing. Instead of searching the entire database the system uses the function to "jump" directly to the related record, minimizing query time.

Q.6. Graph Theory and Networks

Model: Devices are Nodes (Vertices) and communication links are edges.

The Relation:

$R = \{(d_1, d_2) \mid d_1 \text{ can communicate with } d_2\}$

Graph Concept: If communication is bi-directional it is an undirected Graph, if a sensor only sends data to a hub but cannot receive it, Til is a Directed Graph (Digraph).

Ques.7. Symmetric but Not Reflexive

Scenario: The relation "is married to" on the the set of all people.

Symmetric: If person A is married to person B, the Person B is married to Person A.

Not Reflexive. A person cannot be "married to" themselves $(a,a) \notin R$.

Othe Examples: "is a sibling of". You are your sibling's sibling, but you are not your own sibling.